Phase 4: ADHD App Logic and Recommendation System

First we should remember our variables. This table classifies all Phase 4 variables into input, context, and derived categories for the ADHD app. It also defines each variable's data type and summarizes the scientific rationale for including it in cognitive state detection. **I also added the derived variables**

Variable Name	Category	Data Type	Description	Scientific Basis
Stress level	Input	Scale 1-5	User-reported level of overwhelm, pressure, or felt stress in the current moment.	Stress exposure is linked to more persistent and complex ADHD trajectories and greater emotional dysregulation; adults with ADHD often feel hassled and overwhelmed under stress (13.1, 3.5)
Focus/attention	Input	Scale 1-5	User-reported ability to sustain attention on the current task and resist distraction.	ADHD attention is state-dependent and shaped by executive control plus arousal/energetic regulation; low focus may reflect executive dysfunction or dysregulated activation rather than a constant deficit (2.1, 2.2)
Fatigue	Input	Scale 1-5	User-reported mental or physical tiredness that may reduce effort capacity.	Fatigue reduces cognitive resources and is closely tied to impaired attention and executive performance in ADHD-related cognitive functioning (3.3, 16.1)
Sleepiness	Input	Scale 1-5	User-reported daytime drowsiness or tendency to doze during tasks.	Daytime sleepiness is a major driver of attention errors and reaction-time variability in adults with ADHD; on-task sleepiness strongly predicts omission errors (3.2, 3.1)
Motivation	Input	Scale 1-5	User-reported willingness or drive to start and continue goal-directed behavior.	ADHD is associated with reward/motivation deficits and altered dopamine reward-pathway functioning, especially preference for immediate rewards and reduced persistence on unrewarding tasks (5.1, 4.1)
Sleep quality	Input	Scale 1-5	User-reported quality of the previous night's sleep, including restfulness and continuity.	Poor sleep quality is common in ADHD and contributes to attention, mood, and cognitive difficulties; adults with ADHD show substantially worse sleep quality than controls (3.2, 17.1)
Task difficulty	Context	Scale 1-5	Perceived complexity or cognitive demand of the current task.	Higher task difficulty increases cognitive load and taxes limited executive/energetic resources, raising risk of overload or disengagement (2.2, 10.1)
Workload	Context	Scale 1-5	Amount of work pressure, demands, or number of competing tasks currently present.	High workload intensifies mental effort demands and can deplete attention resources; ADHD difficulties are amplified when many demands compete at once (5.1, 6.2)

Current activity	Context	Categorical	What the user is doing right now, such as studying, writing, scrolling, commuting, or resting.	ADHD performance varies strongly by activity type; boring/monotonous activities worsen symptoms, while stimulating or rewarding activities can improve engagement or trigger hyperfocus (5.1, 12.3)
Environment type/state	Context	Categorical	External setting and conditions, such as quiet, noisy, social, distracting, structured, or chaotic.	ADHD symptoms emerge through interaction with environmental demands; job control, social support, and environmental structure significantly affect distress and functioning (13.1, 9.1)
Medications	Context	Categorical	Current medication status, such as on stimulant, off stimulant, non-stimulant, or none.	Medication status changes attention, arousal, and reward processing; methylphenidate can normalize frontal theta activity and aberrant reward responses (3.3, 7.1)
Productive times	Context	Time	User-identified time windows when focus, energy, or productivity is usually best.	ADHD cognition is influenced by circadian timing, arousal, and sleep-wake mismatch; time-of-day effects can alter performance and should be considered in interpretation (3.4, 17.1)
Recurring overload periods	Derived	Time	Time blocks or situations the system learns are repeatedly associated with overload states.	Pattern recognition over time can identify repeated overload/drift states from behavior and self-report; ADHD attention is dynamic and benefits from state-based adaptive support (1.1, 1.4)
Productive time patterns	Derived	Time	Learned patterns showing when the user most often sustains effective attention or completes tasks.	Human-in-the-loop behavioral sensing can infer attention rhythms and personalize support according to fluctuating focus and engagement patterns (1.5, 1.6)
Preferred regulation strategies	Derived	Categorical	Strategies the system learns are most effective for this user, such as breaks, body doubling, music, or step breakdown.	ADHD interventions work best when personalized and reinforced over time; adaptive feedback engines can refine which nudges or coping supports are most helpful for each user (14.1, 1.2)
Sleep-wake patterns	Derived	Time	Learned trends in bedtime, wake time, sleepiness windows, and circadian mismatch.	ADHD is frequently associated with delayed circadian phase, poor sleep efficiency, and repeated sleep-related cognitive impairment, making longitudinal sleep patterning clinically relevant (3.4, 17.1)
Stress triggers	Derived	Categorical	Learned situations, activities, or contexts that repeatedly precede spikes in stress or dysregulation.	Stress and emotional dysregulation in ADHD are context-sensitive and often recur in recognizable patterns tied to demands and coping failures (13.1, 3.5)
Task completion patterns	Derived	Categorical	Learned tendencies in initiation, switching, abandoning, and completing tasks over time.	ADHD involves variability in task initiation, persistence, and completion tied to executive dysfunction, effort allocation, and reward sensitivity; behavioral sensing can detect such patterns for adaptive support (2.1, 1.3)

3. Cognitive State Definitions

Seven primary cognitive states. The full definitions with characteristics, neural mechanisms, and research basis are presented below. The table defines the seven core ADHD-related cognitive states for Phase 4 and links each state to its main behavioral characteristics and scientific basis. It is useful as the foundation for the pattern-recognition layer before building the recommendation logic.

Cognitive State	Definition	Key Characteristics	Primary Neural/Mechanistic Basis	Research Basis
Cognitive Overload	A state in which current task demands exceed the user's available working memory, effort, and executive-control capacity.	High stress or overwhelm; low focus; rising error risk; difficulty holding steps in mind; task feels "too much"; often worsens under high workload or high task difficulty.	Limited working-memory capacity plus dysregulated energetic pools of effort, arousal, and activation; overload taxes executive control and inhibition.	ADHD performance depends on executive control and energetic state; cognitive load and task difficulty can overwhelm limited resources (2.1, 2.2, 10.1)
Motivation Deficit	A low-drive state in which the user struggles to initiate or persist on a task because it feels insufficiently rewarding, too delayed in payoff, or too effortful.	Low motivation; procrastination; task avoidance; preference for immediately rewarding alternatives; better engagement when reward, novelty, or salience increases.	Dopamine reward-pathway dysfunction, especially altered striatal reward processing and delay aversion; weak effort allocation for low-salience tasks.	ADHD is linked to reward/motivation deficits, reduced dopamine markers in reward circuitry, and strong preference for immediate rewards; novelty and reinforcement improve performance (5.1, 6.1, 4.1, 9.1)
Hyperfocus (Productive)	An intense absorption state directed toward a goal-aligned task, where attention is highly concentrated and performance may improve.	Deep concentration; time distortion; reduced awareness of surroundings; sustained effort on meaningful work; often occurs when task is interesting, challenging, and matched to skill.	Extreme allocation of attentional resources driven by intrinsic motivation, salience, and reward engagement; related to flow-like states.	Hyperfocus is defined as intense concentration with diminished perception of unrelated stimuli and is often triggered by interesting tasks; it closely overlaps with flow and can enhance task performance (11.1, 11.2, 12.1)
Hyperfocus (Unproductive)	An intense absorption state directed toward a non-goal-aligned activity, causing neglect of intended priorities despite strong sustained attention.	Locked onto scrolling, gaming, browsing, or side tasks; loss of time awareness; difficult task-switching; intended work is displaced by more rewarding stimulation.	Attention dysregulation plus reward-driven salience capture; highly stimulating activities provide immediate feedback and intrinsic reward that sustain attention away from intended goals.	In ADHD, attention is dysregulated rather than globally weak; highly rewarding, structured media activities can capture attention intensely and pull resources away from broader goals (11.2, 12.3, 12.2, 11.3)

Fatigue-Induced Attention Failure	A state in which sleepiness or fatigue degrades vigilance, sustained attention, and cognitive efficiency.	Moderate to high sleepiness; omission errors; slower or more variable responses; poor concentration; reduced task endurance; often worse at unproductive circadian times.	Daytime sleepiness, poor sleep quality, and hypo-arousal reduce frontoparietal and executive efficiency; EEG slowing and low alertness contribute to attentional lapses.	Adults with ADHD often have poor sleep quality and higher sleepiness; on-task sleepiness strongly predicts omission errors and reaction-time variability (3.2, 3.1, 3.3, 16.1)
Emotional Dysregulation	A state of heightened emotional reactivity in which stress, frustration, irritability, or overwhelm disrupts attention, inhibition, and problem solving.	Irritability; frustration; feeling overwhelmed; reduced cognitive control; stress-sensitive attention collapse; emotional intensity disproportionate to current demand.	Weak top-down regulation from executive networks over bottom-up emotional systems; altered connectivity involving prefrontal control systems and emotion-generation circuits.	ADHD emotional dysregulation involves reactivity, frustration, and impaired regulation linked to executive-function weaknesses and stress exposure (13.1, 3.5)
Executive Dysfunction / Initiation Paralysis	A state in which the user wants to act but cannot effectively plan, sequence, organize, or begin the task.	Trouble starting; cannot break task into steps; disorganization; switching without progress; “stuck” feeling despite intention; poor follow-through.	Impaired goal-directed executive processes involving frontostriatal/frontoparietal systems; reduced inhibitory control, working memory support, and metacognitive regulation.	Executive dysfunction in ADHD affects goal-directed behavior, attention guidance, organization, and initiation; working-memory and organizational problems contribute to in-situ performance failure (2.2, 2.3, 14.1)

4. Cognitive State Detection Table

The detection table maps specific combinations of input and context variables to each cognitive state. Each state has **three variant profiles representing different pathways to the same state**: this ensures the system recognizes that the **same cognitive state can arise from multiple patterns**. This approach is consistent with the heterogeneity inherent in ADHD, where multiple neuropsychological pathways can produce similar functional outcomes. For example, Cognitive Overload can emerge from (a) high stress + low focus + high workload at night, (b) fatigue + studying in a noisy environment, or (c) high stress + low motivation while off medication. Each route produces the same state but through different variable combinations. The table is useful as the core pattern-recognition logic for translating momentary data into interpretable states before generating recommendations.

Cognitive State	Variant	Input Variables	Context Variables	Research Basis
Cognitive Overload	Variant 1	Stress = High; Focus = Low; Fatigue = Moderate	Workload = High; Task difficulty = High; Time = Night	High task demand plus reduced energetic resources can overwhelm executive control, especially under high cognitive load and low attention capacity (2.1, 2.2, 10.1)

Cognitive Overload	Variant 2	Stress = Moderate; Focus = Low; Fatigue = High; Sleepiness = Moderate	Current activity = Studying; Workload = High; Environment = Noisy/distracting	Overload risk rises when working memory is taxed while fatigue and distraction reduce available control resources (10.1, 3.3, 3.1)
Cognitive Overload	Variant 3	Stress = High; Focus = Low; Motivation = Low	Task difficulty = High; Multiple deadlines/high workload; Medications = Off	ADHD performance worsens when high demands meet low motivation and lower regulatory support; effort allocation and executive control both weaken (2.2, 5.1, 7.1)
Motivation Deficit	Variant 1	Motivation = Low; Stress = Low; Focus = Moderate	Current activity = Administrative/boring task; Task difficulty = Moderate; Reward salience = Low	ADHD motivation deficits are especially visible in boring, effortful, low-reward tasks; low salience reduces engagement and persistence (5.1, 6.2, 4.1)
Motivation Deficit	Variant 2	Motivation = Low; Focus = Low; Fatigue = Low	Task difficulty = High; Reward delayed = Yes; Current activity = Long-term project	Delay aversion and preference for immediate reward make long-horizon tasks harder to start or sustain in ADHD (6.1, 9.1)
Motivation Deficit	Variant 3	Motivation = Low; Stress = Moderate; Sleepiness = Low	Workload = Moderate; Environment = Calm; Time = Non-productive period	Even without overload, engagement can collapse when the task lacks reward value or is poorly aligned with the user's motivational profile and circadian rhythm (5.1, 6.2, 17.1)
Hyperfocus (Productive)	Variant 1	Focus = High; Motivation = High; Stress = Low	Current activity = Writing/coding/studying goal-aligned task; Task difficulty = Moderate-High; Goals = Clear	Hyperfocus/flow is more likely when the task is interesting, meaningful, and matched to skill with clear goals and feedback (11.1, 12.1)
Hyperfocus (Productive)	Variant 2	Focus = Very High; Fatigue = Low; Motivation = High	Time = Productive period; Environment = Quiet/structured; Workload = Moderate	Deep sustained attention emerges more easily in supportive contexts that reduce interruptions and align with learned attention rhythms (1.1, 1.4, 11.1)
Hyperfocus (Productive)	Variant 3	Focus = High; Stress = Moderate; Motivation = High	Current activity = Challenging but meaningful task; Feedback = Immediate; Medications = On/effective	Challenging tasks with immediate feedback and adequate regulatory support can channel ADHD attention into productive flow-like concentration (12.1, 12.2, 7.1)
Hyperfocus (Unproductive)	Variant 1	Focus = High; Motivation = High for wrong task; Stress = Moderate	Current activity = Scrolling/gaming/video browsing; Intended task difficulty = High; Reward salience of distraction = High	Highly stimulating media can capture attention through immediate reward and feedback, producing intense but misdirected focus (11.2, 12.2, 11.3)

Hyperfocus (Unproductive)	Variant 2	Focus = Very High; Sleepiness = Low; Motivation = Low for intended task	Environment = Alone; Current activity = Side research/tab switching rabbit hole; Workload = High	ADHD attention can lock onto interesting non-priority material when intended work feels less rewarding, creating attention misallocation rather than simple inattention (12.3, 5.1)
Hyperfocus (Unproductive)	Variant 3	Focus = High; Stress = High; Emotional tension = Moderate	Current activity = Avoidant deep dive into hobby/secondary task; Deadlines = Present; Time blindness = Likely	Hyperfocus can function as escape from aversive or effortful demands, especially when pressure is high and alternative stimulation is more rewarding (11.2, 5.1, 11.3)
Fatigue-Induced Attention Failure	Variant 1	Sleepiness = High; Fatigue = High; Focus = Low	Time = Night; Sleep quality = Poor; Current activity = Studying/reading	Daytime or on-task sleepiness strongly predicts omission errors and vigilance failures in ADHD, especially after poor sleep (3.2, 3.1)
Fatigue-Induced Attention Failure	Variant 2	Fatigue = Moderate; Sleepiness = Moderate; Stress = Moderate	Time = Early morning after late sleep; Environment = Quiet; Workload = Moderate	Circadian mismatch and sleep deprivation reduce alertness and frontoparietal efficiency, worsening attention even in otherwise manageable tasks (3.4, 17.1)
Fatigue-Induced Attention Failure	Variant 3	Sleepiness = High; Focus = Low; Motivation = Low	Medications = Off/withheld; Task difficulty = Low-Moderate; Current activity = Sustained attention task	Hypo-arousal and sleepiness can account for attention failures often attributed to ADHD alone; medication status may modulate this state (3.3, 3.5)
Emotional Dysregulation	Variant 1	Stress = High; Focus = Low; Motivation = Low	Current activity = Conflictual or evaluative task; Workload = High; Environment = Social pressure	ADHD emotional dysregulation is marked by frustration, overwhelm, and stress-sensitive reductions in cognitive control (13.1, 3.5)
Emotional Dysregulation	Variant 2	Stress = High; Fatigue = Moderate; Sleepiness = Moderate	Sleep quality = Poor; Time = Late day; Current activity = Studying/working under deadline	Poor sleep and fatigue amplify emotional reactivity and weaken top-down regulation, increasing the likelihood of overwhelm (16.1, 13.1)
Emotional Dysregulation	Variant 3	Stress = Moderate; Focus = Low; Irritability/frustration = High	Task difficulty = High; Feedback = Negative or delayed; Environment = Distracting	Executive weaknesses and emotion regulation deficits make challenging, frustrating contexts more likely to trigger dysregulation (13.1, 2.2)
Executive Dysfunction / Initiation Paralysis	Variant 1	Motivation = Moderate; Focus = Low; Stress = Moderate	Current activity = About to start large task; Task difficulty = High; Steps = Unclear	Executive dysfunction can impair planning, sequencing, and initiation even when the user wants to act (2.2, 14.1)

Executive Dysfunction / Initiation Paralysis	Variant 2	Motivation = Moderate; Fatigue = Moderate; Focus = Low	Workload = High; Multiple open tasks; Environment = Chaotic	Working-memory and organizational problems create in-situ performance failures when too many tasks compete for control (2.1, 14.1)
Executive Dysfunction / Initiation Paralysis	Variant 3	Motivation = High; Stress = High; Focus = Low	Current activity = Planning/transitioning; Time = Non-productive period; Medications = Off or waning	The user may care about the task but still get stuck when executive control, timing, and arousal regulation are insufficient for action initiation (2.1, 3.4, 7.1)

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5. Recommendation Library

For each cognitive state, the recommendation library provides evidence-based interventions drawn from multiple therapeutic modalities:

- CBT is the primary evidence-supported psychological intervention for adult ADHD, emphasizing environmental restructuring, skill rehearsal, psychoeducation, and structured practice (14.1, 15.1).
- Mindfulness-based CBT (MBCT) has been tested in RCTs and systematic reviews for adult ADHD, showing effects on attention, emotion dysregulation, and quality of life (14.2, 14.3).
- DBT adaptations for ADHD combine psychoeducation, skills training, and validation in group formats (14.4).
- Metacognitive therapy strengthens awareness of cognition and executive strategy use (14.4).
- Sleep/circadian interventions including bright light therapy and sleep hygiene can improve attention and cognitive functioning in ADHD (3.4, 16.1).
- Digital micro-interventions such as DopBoost (mood fuel, zen zest, reward rush, focus ritual), soft nudges, and body doubling have been proposed in adaptive ADHD productivity frameworks (1.5, 1.4).

The table is useful as the action layer that follows state detection, grounded in CBT, mindfulness, DBT, coaching, sleep, and adaptive micro-intervention research.

Cognitive State	Priority Level	Recommended Interventions	Scientific Basis
Cognitive Overload	High	Break task into smaller steps; simplify task scope to the next actionable step; take a short break; reduce distractions; use a brief breathing exercise before restarting	CBT and structured skills training support breaking down tasks, environmental restructuring, and supported implementation for ADHD self-regulation; mindfulness-based approaches support brief attention-reset and stress reduction; overload is tied to limited executive

			and energetic resources, so reducing task demand and restoring regulation is appropriate (14.1, 14.3, 14.5, 2.2)
Motivation Deficit	High	Add micro-rewards; switch to a more engaging subtask; use a timer/Pomodoro; gamify progress; use body doubling; change environment to increase stimulation or accountability	ADHD motivation improves when tasks are salient, novel, rewarded, structured, or socially supported; reinforcement can normalize performance, while coaching/body-doubling and adaptive nudges can increase follow-through and activation (6.1, 6.2, 5.1, 1.5, 1.2)
Hyperfocus (Productive)	Moderate	Set check-in timers for food/water/stretching; protect the focus session from interruptions; send gentle reminders for hydration, posture, and time checks; preserve the current workflow if goal-aligned	Hyperfocus/flow can improve performance when attention is aligned with meaningful work; the best intervention is not interruption but light-touch regulation that protects the state while preventing neglect of basic needs, consistent with soft prompting and nonintrusive adaptive support (11.1, 12.1, 1.4, 1.6)
Hyperfocus (Unproductive)	High	Use a soft redirect prompt; set a transition timer; acknowledge the pull of the activity and then redirect; use a “one more minute, then switch” strategy; surface the intended next task	ADHD support tools should avoid harsh interruption and instead use gentle, context-aware prompts; hyperfocus on highly rewarding distractions reflects attention dysregulation, so low-friction transition cues and reflective nudges are more likely to work than rigid commands (11.2, 11.3, 1.5, 1.4, 1.7)
Fatigue-Induced Attention Failure	High	Recommend sleep hygiene; suggest a short power nap if feasible; prompt brief physical movement; use bright light exposure; switch to a lighter task; defer demanding work until a better alertness window	Sleepiness is a major driver of ADHD attention failure; behavioral sleep interventions and circadian supports improve cognitive and emotional functioning, while matching task demand to alertness state is consistent with sleep/arousal evidence (3.1, 16.1, 3.4, 3.2)
Emotional Dysregulation	High	Use grounding or breathing exercises; insert a mindfulness pause; apply DBT-informed emotion labeling; step away from the triggering context; use a self-compassion prompt before re-engaging	Mindfulness-based CBT and DBT-style skills are evidence-supported for ADHD-related self-regulation and emotion regulation; emotional overwhelm reduces cognitive control, so naming emotions and briefly lowering arousal before task resumption is clinically coherent (14.2, 14.4, 14.3, 13.1)
Executive Dysfunction / Initiation Paralysis	High	Provide the first micro-step; use body doubling; offer a structured checklist; reduce choice paralysis by presenting one next action; prepare the environment before starting	CBT, metacognitive therapy, coaching, and structured skills training support externalizing task structure, reducing ambiguity, and improving initiation; body-doubling and adaptive prompting further help convert intention into action (14.1, 14.6, 14.4, 14.5, 1.5)

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6. Example Walkthrough

Using the example from the original Phase 4 specification:

USER INPUTS: Stress = High, Focus = Low, Fatigue = Moderate → ***3 combinations is enough for the low fidelity prototype but functional (derived variables will be for later)***

CONTEXT VARIABLES: Workload = High, Current activity = Studying, Time = Night

PATTERN RECOGNITION: The scoring function assigns the highest score to Cognitive Overload (high stress contributes +2.0, low focus +2.0, moderate fatigue +1.5, high workload +1.5, night time +1.0 = total ~8.0 → if, and, or, etc). Secondary scores for Emotional Dysregulation and Executive Dysfunction are lower.

IF (stress>4 AND focus < 2 OR stress >3 AND fatigue >4)

THEN

STATE= COGNITIVE OVERLOAD

IF COGNITIVE OVERLOAD IS HAPPENNING

SUGGESTION IS: “VVVV”

INFERRED STATE: Cognitive Overload (Variant 1)

RECOMMENDATIONS:

1. Break the task into 1–2 micro-steps
2. Take a short reset break (3–5 min)
3. Reduce distractions in the environment
4. Try a brief breathing exercise
5. Shrink task scope to only the next action

This matches the user's original example output and is grounded in CBT-based task decomposition and environmental restructuring strategies supported by clinical evidence for adult ADHD (14.1, 14.3).